

Gabriel Freedberg

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Software Engineer

Backend software engineer with production API gateway experience in OAuth2, mTLS, and Kubernetes-based environments. Comfortable owning features end-to-end, from design through client-facing rollout, with a track record of finding and fixing defects, building internal tooling, and improving test coverage.

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, SQL, C++ (academic coursework)

Backend: Spring Boot, Node.js, REST APIs, OAuth2, mTLS, API Gateway development

Infrastructure & Tools: Kubernetes, kubectl, GitLab, AWS, Splunk, GitHub Actions

Databases & Testing: SQL, SQLite, relational database design, automated test generation, regression testing, API validation

AI-Assisted Development: Claude, Kiro, Codex, OpenAI API

EXPERIENCE

Spectrum — Software Engineer II, API Gateway / PINXT May 2024 – Present

- Build and maintain backend features for Spectrum's API Gateway team, supporting OAuth2 authorization flows, client onboarding, API access controls, and gateway behavior across multiple environments.
- Onboarded approximately 30 clients and 7 APIs onto gateway services, translating integration requirements into implementation changes, access configuration, validation steps, and release-ready updates.
- Implemented certificate parsing logic for an mTLS client onboarding flow, extracting Common Name and other required metadata fields for backend authorization.
- Package and ship releases for gateway services, tracking code through development and performance testing, and resolving bugs surfaced anywhere from dev through production.
- Identified 5 gateway code defects, including 2 production issues coordinated to resolution and 3 lower-environment issues fixed directly.
- Designed and developed a data-driven test framework to replace brittle existing tests, generating valid call sequences across client feature configurations.
- Contributed to Spring Boot version upgrades across gateway services, resolving 12 breaking changes to align with current team framework standards.
- Partner with developers, testers, and stakeholders to clarify integration requirements and troubleshoot OAuth2 and mTLS issues through development and testing environments.
- Review approximately 5 merge requests per day and maintain gateway code across BAU work, including API whitelisting, access enablement, debugging, and environment-specific issue resolution.

Spectrum — Software Engineer Intern, API Gateway Summer 2023

- Built internal tooling to retrieve Kubernetes cluster status and write results to a jumpbox-accessible file, helping developers check cluster state before restarts and deployments.
- Created a PKCE / code verifier generation tool allowing developers to generate configurable verifier/challenge pairs for repeatable testing.
- Updated gateway code paths used in development testing and deployment workflows, gaining hands-on experience with production-oriented API gateway practices.

PROJECTS

Daily History Notifier | Python, JavaScript, GitHub Actions, Wikimedia API, Spotify API, OpenAI API

- Built a daily learning app that selects a historical event from Wikimedia's "On this day" feed, retrieves related Wikipedia content, finds relevant Spotify podcast episodes, and formats a user-facing digest.
- Integrated Wikimedia, Spotify, OpenAI, GitHub Actions, and webhook delivery into a complete workflow, with local fallback logic when API keys or quota are unavailable.
- Automated delivery through GitHub Actions and cron scheduling, with support for Discord, Slack, generic webhooks, GitHub Issues, and terminal output.

Local Deal Alert Platform | Python, JavaScript, SQLite, Supabase, SMTP, Auth, Regression Testing

- Built a local-first deal alert prototype with separate shopper and business account flows, including favorites, deal search, personalized alerts, and expiration-aware deal states.
- Added prototype authentication flows including account creation, login, password reset, hashed local sessions, and session revocation.
- Designed a migration path to hosted Supabase-backed auth and data storage, including checked-in schema and row-level security policy files, with regression coverage for auth, CRUD, and matching logic.

EDUCATION

University of Colorado Boulder — B.S. Computer Science, Minor in Computer Engineering | May 2024 | GPA: 3.8/4.0 | Dean's List